

# Add Stones, Get Crayfish – Is it that Simple?

STEIN I. JOHNSEN <sup>1,†</sup> AND TROND TAUGBØL <sup>2</sup>

<sup>1</sup>Norwegian Institute for Nature Research, Fakkeltgården, NO-2624 Lillehammer, Norway

<sup>2</sup>Glommen's and Laagen's Water Management Association, NO-2605 Lillehammer, Norway

<sup>†</sup>Corresponding Author — E-mail: [stein.ivar.johnsen@nina.no](mailto:stein.ivar.johnsen@nina.no)

**Abstract.**— In an attempt to increase the abundance of noble crayfish, *Astacus astacus* (Linnaeus), in a soft-bottomed area of lake, we modified the biotope by adding 1000 tonnes of natural stones. The effect on crayfish abundance in both the biotope adjustment site and a control site was monitored by diving and baited traps over a period of 13 years. The study lake harbours a good, harvestable noble crayfish population, but in some areas the crayfish density is very low, probably due to lack of shelter. The abundance of noble crayfish at the biotope adjustment site increased significantly over the years, whereas the abundance at the control site did not display similar changes. At the end of the study period the abundance of crayfish at the biotope adjustment site had reached a level of great harvesting interest. The study shows that, with relatively simple methods, it is possible to increase abundance and yield of noble crayfish by increasing shelter availability. [Keywords:— abundance; *Astacus astacus*; biotope adjustment; shelter; stones].

## INTRODUCTION

Crayfish can be found in a wide range of biotopes, from small ponds to large lakes, streams to large rivers, cave pools and estuaries. The abundance of crayfish often depends on shelter availability (e.g., Capelli and Magnuson 1983; Lodge and Hill 1994; Nyström et al. 2006) due to risk of predation from predatory fish and terrestrial predators like mammals and birds (Fleury and Sherry 1995; Beja 1996; Nyström et al. 2006). In areas with soft or muddy sediments, crayfish are less abundant (e.g., Kirjavainen and Westman 1999). Among crayfish species regarded as non-burrowers, e.g., cool-water crayfish like the noble crayfish *Astacus astacus* (Linnaeus), and *Pacifastacus leniusculus* (Dana), their abundance is positively correlated to abundance of physical shelters/refuges (Lodge and Hill 1994; Nyström et al. 2006).

Small crayfish are found more often in shallow littoral areas with abundant shelters than large adult crayfish (Skurdal et al. 1988; Englund and Krupa 2000). Although larger crayfish may be less susceptible to predation (Englund and Krupa 2000), shelters are still important as large crayfish are vulnerable in their soft-shell period after moulting (Ackefors 1996). As one of several management strategies, shelter additions were proposed by Lodge and Hill (1994) to increase yield of cool-water crayfish.

There are unique social, cultural, and local commercial values connected to crayfish fisheries, especially in Scandinavia (Taugbøl and Skurdal 1993; Swahn 2004). In addition, the native noble crayfish is a red-listed species (IUCN 2006; Kålås et al. 2006) threatened by the human-facilitated spread of plague-carrying, non-indigenous crayfish species. The possibility for exploitation and local economic benefits may be important for the conservation of the species, as local people who benefit from the noble crayfish are inclined to protect against illegal stocking of non-indigenous crayfish (Taugbøl 2004). Hence, measures to strengthen populations

of noble crayfish, increase the yield, and encourage harvesting are of great interest both for exploitation and conservation purposes.

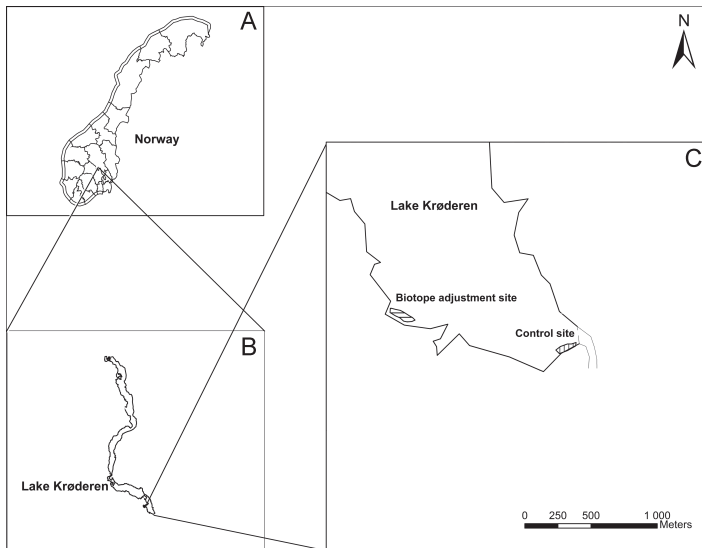
The present study was carried out in a large Norwegian lake, Lake Krøderen. There is a significant noble crayfish fishery in the lake, but in many areas crayfish are almost absent, and no harvesting effort takes place. It was hypothesised that differences in shelter availability was the reason for the uneven distribution of crayfish in the lake. A biotope adjustment project was started in 1990, with the aim of increasing crayfish abundance in an area where crayfish were almost absent. The project involved placing natural stones as shelter for crayfish. Even though the importance of shelter on crayfish abundance is well known (e.g., Capelli and Magnuson 1983; Lodge and Hill 1994; Nyström et al. 2006), few studies document the effect of adjusting crayfish biotopes by adding stones as shelters/refuges (but see Huolila et al. 1997; Taugbøl 2005). In this paper, we report the effect of such a biotope adjustment project on noble crayfish abundance over a period of 13 years.

## METHODS

### Study Site

Lake Krøderen (133 m above sea level) is situated in the southern part of Norway (Figure 1) and has a surface area of 43.8 km<sup>2</sup>. The lake is inhabited by noble crayfish, which were introduced in the early 1960s. The fish community in the lake consists of perch (*Perca fluviatilis* Linnaeus), pike (*Esox lucius* Linnaeus), brown trout (*Salmo trutta* Linnaeus), arctic char (*Salvelinus alpinus* Linnaeus), whitefish (*Coregonus lavaretus* Linnaeus), minnow (*Phoxinus phoxinus* Linnaeus) and crucian carp (*Carrasius carrasius* Linnaeus).

The study was conducted in the southern part of the lake (Figure 1), at two different sites; the biotope adjustment site where



**Figure 1.** A, map of Norway, B, Lake Krøderen and C, the location of the biotope adjustment and control sites. Permission through Norge Digitalt.

local fishermen seldom fished due to very low catches of crayfish, and a control site offering stable and good catches. The substratum at the biotope adjustment site (0 – 4 m depth) was a typical soft-bottom substrate, covered with quillwort (*Isoëtes* sp.), shoreweed (*Plantago* sp.) and some water-milfoil (*Myriophyllum* sp.). At the control site the substratum was covered with stones providing good shelter for crayfish. Aquatic macrophytes were almost absent at this site.

#### Biotope Adjustment Measures

In 1991 and 1993 approximately 50 and 950 tonnes of rounded natural stones (size 5 – 30 cm) were placed by truck, in winter, on the ice at the biotope adjustment site. Water depth was 2 – 4 m, and after sinking (ice melting) the stones formed piles of 1 – 1.5 m in height. In this survey we monitored one pile with a total area of approximately 100 m<sup>2</sup>.

#### Crayfish Surveys

The development of the crayfish population was monitored over time by diving-surveys and use of baited traps. During diving, the rock formation was searched, rocks turned and visible crayfish caught by hand. The diving (20 – 40 minutes of searching) was carried out in September/October each year in the periods 1990 – 1993 and 1996 – 1998 at the biotope adjustment site and in 2003 at both sites. An estimate of relative density is given as catch per hour diving (CPUE-dive). Trap catches were carried out in early August at both the biotope adjustment site (21 – 64 traps night<sup>-1</sup>) and the control site (50 – 100 traps night<sup>-1</sup>) in 1990 – 1993 and in 1996. An estimate of relative density is given as catch per trap night (CPUE-trap).

#### Statistical Analyses

The effect of changes in abundance over time was tested using linear regression. To test if changes in crayfish abundance at the control site showed a similar trend as at the biotope adjustment site Spearman Rank Order Correlation Coefficients were calculated.

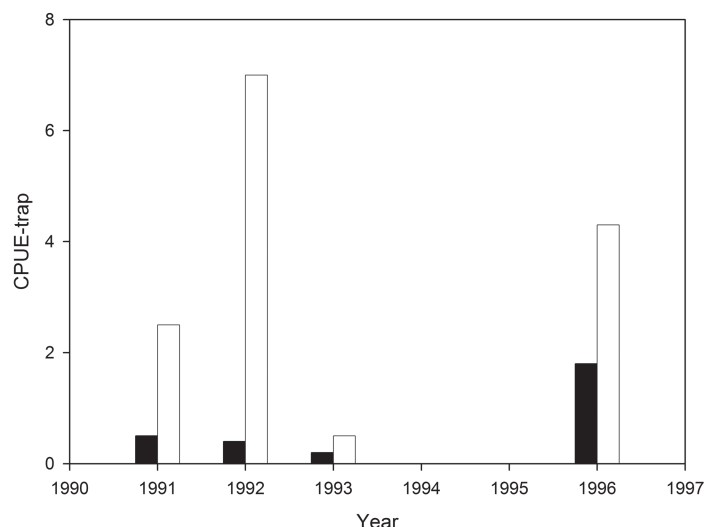
## RESULTS

Before the stones were added, no crayfish were observed when diving at the biotope adjustment site. After the first modest placing of stones in 1991, a few crayfish were recorded (Table 1). There was a significant increase in CPUE-dive throughout the period 1990 – 2003 ( $R^2 = 0.94$ ,  $F_{1,6} = 101$ , 32,  $P < 0.001$  by linear regression) (Table 1). The size of crayfish ranged from 3 – 11 cm carapace length (Table 1). Crayfish abundance at the control site did not display a similar development as at the biotope adjustment site ( $r = 0.4$ ,  $P = 0.6$ ,  $n = 4$ , by Spearman's Correlation) (CPUE-trap data from 1991 – 1993 and 1996) (Figure 2). CPUE-dive data in 2003 gave almost the same results at both sites, with 129 and 108 crayfish in the biotope adjustment site and control site respectively.

## DISCUSSION

This study demonstrates that improving crayfish biotopes by increasing the abundance of shelter may lead to increased crayfish production in previously barren areas. The effects of shelter availability on crayfish abundance have been well documented in previous studies (e.g., Capelli and Magnuson 1983; Lodge and Hill 1994; Nyström et al. 2006), however few studies have actually manipulated crayfish habitats by adding stones (but see Huolila et al. 1997; Taugbøl 2005), and monitored the effects of the manipulations over extended periods of time.

In a dredged river in Finland, Huolila et al. (1997) reported increased density of noble crayfish after supplying artificial shelters in areas with few refuges. However, that study was conducted over a short time period (one year), and only revealed the potential of biotope adjustment by adding stones. However, the duration of the study did not allow long term evaluation of the biotope adjustment on a newly established crayfish population. In another study, Taugbøl (2005) monitored the development of noble crayfish over a longer period of time after biotope adjustments (placing of rock groups, digging out pools etc.) in two canalised rivers in Norway. The biotope adjustments in that study gave no effects on



**Figure 2.** Catch per unit effort (number of noble crayfish caught per trap night) at the biotope adjustment site (black bars) and control site (white bars) in Lake Krøderen in the years 1991 – 1993 and 1996.

**Table 1.** CPUE (number of crayfish caught per hour of diving), total number of crayfish caught per dive and mean (minimum and maximum) carapace length (cm) of noble crayfish caught at the biotope adjustment site in the years 1990 – 1993, 1996 – 1998 and 2003 in Lake Krøderen. The arrows indicate years when stones were added during winter.

| Year   | CPUE | Number of crayfish caught | Mean size (min – max) cm |
|--------|------|---------------------------|--------------------------|
| 1990   | 0    | 0                         | —                        |
| ► 1991 | 8    | 4                         | 9.5 (9 – 10)             |
| 1992   | 21   | 14                        | 6.8 (3 – 11)             |
| ► 1993 | 14   | 9                         | 6.6 (5 – 11)             |
| 1996   | 48   | 24                        | 7.1 (4 – 10)             |
| 1997   | 90   | 30                        | 7.7 (3 – 10)             |
| 1998   | 72   | 35                        | 6.9 (3 – 10)             |
| 2003   | 129  | 68                        | 7.1 (4 – 11)             |

establishment of crayfish or increased crayfish abundance. The study by Taugbøl (2005) highlights the importance of knowing the limitations of crayfish production before implementing measures to manage or alter populations.

The studies of both Huolila et al. (1997) and Taugbøl (2005) included stocking of crayfish after the placing of the stones, whereas this study exclusively depended on natural colonization of crayfish dispersing into, and recruitment within, the biotope adjustment site. In the present study there was an immediate response to the biotope adjustment, displayed by a significant increase in crayfish abundance.

Success in the competition for shelter among crayfish is positively correlated with the size of the crayfish (Ranta and Lindstøm 1992; Ranta and Lindstøm 1993). Limited shelter opportunities may also be worse for small juvenile crayfish than for adults, because they are more susceptible to predation by fish (DiDonato and Lodge 1993). They may also be more susceptible to cannibalism due to more frequent moulting. Figiel et al. (1991) showed that increased shelter abundance probably reduced cannibalism and increased survival in young of the year *Procambarus clarkii* (Girard), even though there were no predators and an excess of food. Increased abundance of shelter will therefore lead to an increased local and overall abundance of crayfish, since the susceptibility to predation and cannibalism is reduced, and survival will be increased in smaller and inferior crayfish.

After 13 years, the crayfish abundance at the biotope adjustment site was at the same level as the control site; i.e., a density offering good catches for the local fishermen. The results from this study clearly show the possibility of locally increasing crayfish production by providing additional stones as refuges. Thus, it may be that simple to increase the abundance of crayfish in an area if the limitation is a shortage of shelters.

#### ACKNOWLEDGMENTS

We wish to thank Dr. Jon Museth and Marte Qvenild for commenting on earlier drafts and Oddgeir Andersen for field assistance. We also wish to thank Margrethe Tingstad for improving the language. Financial support was given by the Norwegian Directorate for Nature Management, Norges Skogeierforbund

and the County governor of Buskerud. In addition, the Norwegian Institute for Nature Research (NINA) has given financial support for the writing of this paper. We also thank the two anonymous reviewers for helpful comments on the manuscript.

#### REFERENCES

- ACKEFORS H (1996). The development of crayfish culture in Sweden during the last decade. *Freshwater Crayfish* 11:627–654.
- BEJA PR (1996). An analysis of otter *Lutra lutra* predation on introduced American crayfish *Procambarus clarkii* in Iberian streams. *Journal of Applied Ecology* 33(5):1156–1170.
- CAPELLI GM AND MAGNUSON JJ (1983). Morphoedaphic and biogeographic analysis of crayfish distribution in northern Wisconsin. *Journal of Crustacean Biology* 3(4):548–564.
- DiDONATO GT AND LODGE DM (1993). Species replacement among *Orconectes* crayfishes in Wisconsin lakes: The role of predation by fish. *Canadian Journal of Fisheries and Aquatic Sciences* 50(7):1484–1488.
- ENGLUND G AND KRUPA JJ (2000). Habitat use by crayfish in stream pools: Influence of predators, depth and body size. *Freshwater Biology* 43(1):75–83.
- FIGIEL CR, BABB JG AND PAYNE JF (1991). Population regulation in young of the year crayfish, *Procambarus clarkii* (Girard, 1852) (Decapoda, Cambaridae). *Crustaceana* 61(3):301–307.
- FLEURY BE AND SHERRY TW (1995). Long-term population trends of colonial wading birds in the southern United States: The impact of crayfish aquaculture on Louisiana populations. *The Auk* 112(3):613–632.
- HUOLILA M, MARJOMÄKI TJ AND LAUKKANEN E (1997). The success of crayfish stocking in a dredged river with and without artificial shelter increase. *Fisheries Research* 32(2):185–189.
- IUCN (2006). Redlist of threatened species. <http://www.iucnredlist.org>.
- KÅLÅS JA, VIKEN Å AND BAKKEN T (eds.) (2006). Norsk Rødliste 2006 – 2006 Norwegian Red List. *Artsdatabanken Norway* [in Norwegian].
- KIRJAVAINEN J AND WESTMAN K (1999). Natural history and development of the introduced signal crayfish, *Pacifastacus leniusculus*, in a small, isolated Finnish lake, from 1968 to 1993. *Aquatic Living Resources* 12(6):387–401.
- LODGE DM AND HILL AM (1994). Factors governing species composition, population size and productivity of cool-water crayfishes. *Nordic Journal of Freshwater Research* 69:111–136.
- NYSTROM P, STENROTH P, HOLMQVIST N, BERGLUND O, LARSON PER AND GRANLÉ W (2006). Crayfish in lakes and streams: Individual and population responses to predation, productivity and substratum availability. *Freshwater Biology* 51(11):2096–2113.
- RANTA E AND LINDSTRÖM K (1992). Power to hold sheltering burrows by juveniles of the signal crayfish, *Pacifastacus leniusculus*. *Ethology* 92:217–226.

- RANTA E AND LINDSTRÖM K (1993). Body size and shelter possession in mature signal crayfish, *Pacifastacus leniusculus*. *Annales Zoologici Fennici* 30:125–132.
- SKURDAL J, FJELD E, HESSEN DO, TAUGBØL T AND DEHLI E (1988). Depth distribution, habitat segregation and feeding of the crayfish *Astacus astacus* in Lake Steinsfjorden. *Nordic Journal of Freshwater Research* 64:113–119.
- SWAHN J-Ö (2004). The cultural history of crayfish. *Bulletin Français de la Pêche et de la Pisciculture* 372–373:243–251.
- TAUGBØL T (2004). Exploitation is a prerequisite for conservation of *Astacus astacus*. *Bulletin Français de la Pêche et de la Pisciculture* 372–373:275–279.
- TAUGBØL T (2005). Biotoptiltak for kreps i Starelva, Stange kommune og Smalelva, Spydeberg kommune: Har de hatt noen effekt? *NINA Rapport* 26. [in Norwegian].
- TAUGBØL T AND SKURDAL J (1993). Noble crayfish catching in Norway: Legislation and yield. *Freshwater Crayfish* 9:134–143.